

WONSAN, Korea, Democratic People's Republic of

WMO: 470550

Lat: 39.183N

Lon: 127.433E

Elev: 36

StdP: 100.89

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-8.4	-25.5	0.4	-6.8	-22.7	0.5	-5.6	9.3	-7.0	8.3	-5.6	3.2	230	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	32.2	23.8	30.5	23.3	28.9	22.7	25.8	29.7	25.1	28.5	24.4	27.5	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	19.9	27.7	24.1	19.1	26.9	23.4	18.3	26.4	80.2	30.0	77.0	28.7	74.2	27.5	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 8.0	6.7	5.6														
(5)			DB	-13.3	34.5	2.6	1.4	-15.2	35.5	-16.7	36.3	-18.1	37.1	-20.0	38.2	
			WB	-14.8	27.3	2.2	1.0	-16.4	28.0	-17.6	28.6	-18.9	29.2	-20.5	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-1.7	0.3	5.4	11.6	16.7	20.2	23.7	24.0	19.6	14.3	7.1	0.8
	DBStd	9.59	3.57	3.42	3.98	4.14	3.96	3.20	3.37	2.61	2.54	3.04	4.14	3.82
	HDD10.0	1215	363	271	154	29	1	0	0	0	4	106	287	
	HDD18.3	2854	622	504	402	204	81	15	1	15	127	338	545	
	CDD10.0	1907	0	0	10	78	208	306	424	436	288	139	18	0
	CDD18.3	504	0	0	0	3	29	72	167	177	53	3	0	0
	CDH23.3	3347	0	0	0	27	238	449	1264	1182	181	6	0	0
	CDH26.7	968	0	0	0	3	54	139	437	309	26	0	0	0
(14) Wind	WSAvg	2.0	2.2	2.2	2.3	2.2	1.9	1.7	1.7	1.6	1.6	1.8	2.1	2.3
(15) Precipitation	PrecAvg	1374	31	33	37	67	100	146	322	278	187	74	71	37
	PrecMax	2048	110	131	100	160	269	361	860	645	436	386	604	187
	PrecMin	709	1	2	4	5	15	37	25	44	21	7	7	0
	PrecStd	246	26	29	28	41	65	98	159	132	110	80	115	45
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	11.5	19.9	26.3	30.8	33.0	34.2	33.8	29.4	24.1	19.5	11.7
		MCWB	3.4	5.5	11.2	14.6	17.8	20.2	24.8	25.6	21.8	16.0	12.1	6.5
	2%	DB	6.4	9.0	16.5	23.5	28.1	30.2	32.4	31.6	27.1	22.0	17.0	9.4
		MCWB	1.9	3.3	8.3	12.8	16.6	20.0	24.6	25.2	20.8	15.5	10.7	4.9
	5%	DB	4.8	7.2	14.1	21.2	26.1	28.0	30.9	29.7	25.2	20.6	14.9	7.7
		MCWB	0.8	2.6	6.8	11.6	15.7	20.0	24.2	24.5	20.3	15.2	9.4	3.7
	10%	DB	3.4	5.5	11.9	19.0	23.9	25.8	29.2	28.2	23.9	19.3	13.2	6.2
		MCWB	-0.2	1.5	5.8	10.7	15.0	19.3	23.7	24.1	20.1	14.7	8.8	2.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	6.3	11.9	16.5	20.1	23.4	26.9	27.0	23.9	18.9	13.7	7.6
		MCDB	7.5	10.4	19.0	22.7	26.6	28.5	31.4	30.8	27.1	21.2	17.3	10.8
	2%	WB	2.5	4.2	9.4	14.4	18.1	22.1	25.7	26.1	22.6	17.5	12.0	5.6
		MCDB	5.4	7.8	15.1	21.1	25.6	26.4	30.0	29.8	25.1	20.3	15.4	8.6
	5%	WB	1.2	2.9	7.5	13.0	17.1	21.2	25.0	25.4	21.6	16.5	10.7	4.1
		MCDB	3.9	6.4	12.7	19.0	23.3	25.2	29.0	28.6	23.9	19.3	13.6	7.2
	10%	WB	0.3	1.8	6.1	11.4	16.1	20.5	24.4	24.7	20.7	15.5	9.4	2.9
		MCDB	2.7	4.9	11.0	17.5	21.3	24.1	28.1	27.5	23.0	18.2	12.6	5.6
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.7	7.3	8.4	8.2	6.3	5.2	5.2	6.4	7.4	6.9	6.6
		MCDBR	7.8	8.7	10.8	12.8	13.0	11.2	8.7	8.0	8.6	9.2	9.2	7.7
		MCWBR	5.7	5.6	5.8	6.2	5.7	5.0	3.8	3.8	4.9	5.5	6.0	5.7
	5% WB	MCDBR	6.8	7.8	9.7	11.4	10.7	8.4	7.0	6.3	6.6	7.3	7.6	7.1
		MCWBR	5.4	5.4	6.1	6.5	5.7	4.5	3.6	3.6	4.5	5.3	5.8	5.7
(40) Clear-Sky Solar Irradiance	taub		0.393	0.458	0.532	0.601	0.613	0.671	0.631	0.547	0.485	0.492	0.452	0.394
	taud		2.014	1.881	1.767	1.676	1.702	1.650	1.794	2.002	2.091	1.945	1.972	2.048
	Ebn at Noon		729	729	718	700	704	663	686	737	758	689	657	694
	Edh at Noon		125	164	204	238	236	249	214	170	146	153	131	113
(44) All-Sky Solar Radiation	RadAvg		2.44	3.31	4.27	4.96	5.42	5.00	4.09	4.13	3.97	3.39	2.35	2.15
	RadStd		0.20	0.26	0.41	0.45	0.56	0.50	0.60	0.46	0.44	0.28	0.32	0.22

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+68

Nomenclature: See separate page